STAT 6410
Homework

Sarah Kilgard

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1. Chapter 5, Problem 1
   Verify that if μ_1 and μ_2 are finite measures on $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$, respectively, then $\mu_{12}(\cdot)$ and $\mu_{21}(\cdot)$, defined by (1.4) and (1.5), respectively, are measures on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$.
   (**Hint**: Use the MCT.)
   Note:

$$\mu_{12}(A) = \int_{\Omega_1} \mu_2(A_{1\omega_1})\mu_1 d(\omega_1) \quad \mu_{21}(A) = \int_{\Omega_2} \mu_1(A_{2\omega_2})\mu_2 d(\omega_2)$$

Ans: WTS $\mu_{12}(\cdot)$ is a measure on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$.

- $\mu_2(\emptyset) = \int_{\Omega_1} \mu_2(\emptyset_{1\omega_1}\mu_1 d(\omega_1) = \int_{\Omega_1} \mu_2(\emptyset) d\mu_1(\omega_1) = 0$
- Since $\mu_2(A_{1\omega_1})$ is a nonnegative measurable function, the integral of it will also be nonnegative
- Define $D \in \mathcal{F}_1 \otimes \mathcal{F}_2$. Define $\{A_n\}_{n\geq 1}$ to be a disjoint sequence of sets in $\mathcal{F}_1 \otimes \mathcal{F}_2$ such that $D = \cup_{n\geq 1} A_n$

  $$\mu_{12}(D) = \int_{\Omega_1} \mu_2(D_{1\omega_1})\mu_1 d(\omega_1)$$

  From Prop 5.1.1, we know for each $\omega_1 \in \Omega_1, A_{n1\omega_1} \in \mathcal{F}_2$

  $$\therefore \ \mu_{12}(D) = \int_{\Omega_1} \mu_2(\cup_{n\geq 1} A_{n1\omega_1})\mu_1 d(\omega_1)$$

  μ_2 is a measure and, $\{A_{n1\omega_1}\}_{n\geq 1}$ is disjoint,

  $$\mu_{12}(D) = \int_{\Omega_1} \sum_{n\geq 1} \mu_2(A_{n1\omega_1})\mu_1 d(\omega_1)$$

  For each n, $\mu_2(A_{n1\omega_1}$ is a nonnegative measurable function, so by MCT:

  $$\mu_{12}(D) = \int_{\Omega_1} \mu_2(\cup_{n\geq 1} A_{n1\omega_1})\mu_1 d(\omega_1) = \sum_{n\geq 1} \int_{\Omega_1} \mu_2(A_{n1\omega_1})\mu_1 d(\omega_1)$$

Thus, $\mu_{12}(\cdot)$ is a measure on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$
In the same way, we can also show that $\mu_{21}(\cdot)$ is a measure on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$

2. Chapter 5, Problem 6
Let $(\Omega, \mathcal{F}, \mu)$ be a σ-finite measure space and f be a nonnegative measurable function. Then, $\int_\Omega f d\mu = \int_{[0,\infty)} \mu(\{f \geq t\}) dt$.
(**Hint:** Consider the product space of $(\Omega, \mathcal{F}, \mu)$ with $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), m)$ and apply Tonelli's theorem to the function $g(\omega, t) = I(f(\omega) \geq t)$, after showing that g is $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$-measurable.)

Ans:

Proof. **WTS:** g is $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$-measurable

$g(\omega, t) = I(f(\omega) \geq t) = I_{\{(\omega,t) \in \Omega \times \mathbb{R}_+ : f(\omega) \geq t\}}(\omega, t)$

$f(\omega)$ is $\mathcal{F}$ measurable and t is $\mathcal{B}(\mathbb{R}_+)$ measurable, so $\{(\omega, t) \in \Omega \times \mathbb{R}_+ : f(\omega) \geq t\} \in \mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$.

Thus, $g(\omega, t)$ is $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$- measurable, since it is an indicator function for a $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$-measurable set.

Clearly, $g(\omega, t)$ is a nonnegative function as it can only take on the values 0 and 1. Thus, we can use Tonelli's theorem.

$$k_1(\omega) = \int_{\mathbb{R}_+} g(\omega, t) m(dt) = \int_{\mathbb{R}_+} I(f(\omega) \geq t) m(dt) = m(\{(\omega, t) \in \Omega \times \mathbb{R}_+ : f(\omega) \geq t\}) = f$$

$$k_2(t) = \int_\Omega g(\omega, t) \mu(d\omega) = \int_\Omega I(f(\omega) \geq t)) \mu(d\omega) = \mu(\{f \geq t\})$$

From Tonelli's Theorem, we know that $\int_\Omega k_1 d\mu = \int_{\mathbb{R}_+} k_2 dm = \int_{[0,\infty)] k_2(t) dt}$

$$\therefore \int_\Omega f d\mu = \int_{[0,\infty)]} \mu(f(\omega) \geq t) dt$$

□

3. Chapter 5, Problem 10
Show that $I \equiv \int_0^\infty e^{-x^2/2} dx = \sqrt{\frac{\pi}{2}}$ (**Hint**: By Tonelli's theorem, $I^2 = \int_0^\infty e^{-\frac{(x^2+y^2)}{2}} dxdy$. Now use the change of variables $x = r\cos\theta$, $y = r\sin\theta$, $0 < r < \infty$, $0 < \theta < \frac{\pi}{2}$.)

Ans:

$$I = \int_0^\infty e^{-x^2/2}dx = \int_0^\infty e^{-y^2/2}dy$$

By Tonelli's theorem:$I^2 = \int_0^\infty e^{-\frac{(x^2+y^2)}{2}}dxdy$

Now we will use the change of variables where:$x = r\cos(\theta)$ & $y = r\sin(\theta)$

$I^2 = \int_0^\infty \int_0^{\frac{\pi}{2}} e^{-((r\cos(\theta))^2+(r\sin(\theta))^2)/2}Jd\theta dr$, where J is the Jacobian

$$J = \frac{d}{dr}r\cos(\theta)\frac{d}{d\theta}r\sin(\theta) - \frac{d}{d\theta}r\cos(\theta)\frac{d}{dr}r\sin(\theta) = r$$

$$I^2 = \int_0^\infty \int_0^{\frac{\pi}{2}} e^{-(r^2(\cos(\theta)^2+\sin(\theta)^2)/2}rd\theta dr = \int_0^\infty \int_0^{\frac{\pi}{2}} e^{-r^2/2}rd\theta dr$$

$$= \int_0^\infty \left[e^{-r^2/2}r\left(\frac{\pi}{2}\right) - e^{-r^2/2}r(0)\right]dr$$

$$= \frac{\pi}{2}\int_0^\infty re^{-r^2/2}dr$$

$$\frac{d}{dr}\left(-e^{-r^2/2}\right) = re^{-r^2/2}$$

$$I^2 = \frac{\pi}{2}\int_0^\infty \frac{d}{dr}\left(-e^{-r^2/2}\right)dr = \frac{\pi}{2}\left[-e^{-r^2/2}\right]_0^\infty = \frac{\pi}{2}(0 - -1) = \frac{\pi}{2}$$

$$\therefore\ I = \sqrt{I^2} = \sqrt{\frac{\pi}{2}}$$

4. Chapter 5, Problem 11
Let μ be a finite measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $f, g : \mathbb{R} \to \mathbb{R}_+$ nondecreasing. Show that

$$\mu(\mathbb{R}) \int fg d\mu \geq \left(\int f d\mu \right) \left(\int g d\mu \right)$$

(**Hint**: Consider $h(x_1, x_2) = (f(x_1) - f(x_2))(g(x_1) - g(x_2))$ on $\mathbb{R}^2$ and integrate w.r.t. $\mu \times \mu$.)

Because f and g are nondecreasing,

If $x_1 > x_2 \to (f(x_1)-f(x_2)) \geq 0$, $(g(x_1)-g(x_2)) \geq 0 \Rightarrow (f(x_1)-f(x_2))(g(x_1)-g(x_2)) \geq 0$

If $x_1 \leq x_2 \to (f(x_1)-f(x_2)) \leq 0$, $(g(x_1)-g(x_2)) \leq 0 \Rightarrow (f(x_1)-f(x_2))(g(x_1)-g(x_2)) \geq 0$

$\Rightarrow h$ is nonnegative $\Rightarrow \int_{\mathbb{R}^2} h(x_1, x_2) d(\mu \times \mu) \geq 0$

$$\int_{\mathbb{R}^2} h(x_1, x_2)(\mu \times \mu)(d(x_1, x_2)) = \int_{\mathbb{R}^2} f(x_1)g(x_1) - f(x_1)g(x_2) - f(x_2)g(x_1) + f(x_2)g(x_2) \, (\mu \times \mu) d(x_1, x_2)$$

$$= \int_{\mathbb{R}^2} f(x_1)g(x_1) d\mu \times \mu - \int_{\mathbb{R}^2} f(x_1)g(x_2) d\mu \times \mu - \int_{\mathbb{R}^2} f(x_2)g(x_1) d\mu \times \mu + \int_{\mathbb{R}^2} f(x_2)g(x_2) d\mu \times \mu$$

applying Tonelli's thm

$$= \int fg d\mu - \int_{\mathbb{R}^2} f(x_1)g(x_2)(\mu \times \mu) d(x_1, x_2) - \int_{\mathbb{R}^2} f(x_2)g(x_1)(\mu \times \mu) d(x_1, x_2) + \int fg d\mu$$

$$= 2 \int fg d\mu - \int_{\mathbb{R}^2} f(x_1)g(x_2)(\mu \times \mu) d(x_1, x_2) - \int_{\mathbb{R}^2} f(x_2)g(x_1)(\mu \times \mu) d(x_1, x_2)$$

$$\Rightarrow \mu(\mathbb{R}) \int_{\mathbb{R}} fg \, d\mu \geq \int_{\mathbb{R}} f d\mu \int_{\mathbb{R}} g d\mu$$

5. Problem A

Consider the following measure spaces
$(\Omega_i = \mathbb{N}, \mathcal{F}_i = \mathcal{P}(\mathbb{N}),\ \mu_i = \text{counting measure}),\ i = 1, 2.$
Define $f : (\Omega_1 \times \Omega_2) \to \mathbb{R}$ as

$$f(\omega_1, \omega_2) = \mathbf{I}_{\{(n,n):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) - \mathbf{I}_{\{(n,n+1):\, n\in\mathbb{N}\}}(\omega_1, \omega_2)$$

Then,

(i) Prove $\mathcal{F}_1 \otimes \mathcal{F}_2 = \mathcal{P}(\mathbb{N} \times \mathbb{N})$.

(ii) Is f a $\mathcal{F}_1 \otimes \mathcal{F}_2$ measurable function?

(iii) Is f integrable w.r.t $\mu_1 \times \mu_2$?

(iv) Compute the two iterated integrals:

$$\int_{\Omega_1}\left[\int_{\Omega_2} f(\omega_1, \omega_2)d\mu_2(\omega_2)\right]d\mu_1(\omega_1), \quad \int_{\Omega_2}\left[\int_{\Omega_1} f(\omega_1, \omega_2)d\mu_1(\omega_1)\right]d\mu_2(\omega_2)$$

Ans: (i)

Proof. **WTS:** $\mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N}) \subseteq \mathcal{P}(\mathbb{N} \times \mathbb{N})$
Both are σ-algebra on $\mathbb{N} \times \mathbb{N}$, and $\mathcal{P}(\mathbb{N} \times \mathbb{N})$ is the largest σ-algebra on $\mathbb{N} \times \mathbb{N}$.
Thus: $\mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N}) \subseteq \mathcal{P}(\mathbb{N} \times \mathbb{N})$

WTS: $\mathcal{P}(\mathbb{N} \times \mathbb{N}) \subseteq \mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N})$
$\mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N}) = \sigma\langle\{A_1 \times A_2 : A_1, A_2 \in \mathcal{P}(\mathbb{N})\}\rangle$

Suppose $A \in \mathcal{P}(\mathbb{N} \times \mathbb{N})$. Thus, there exist $a_1, a_2, ... \in \mathbb{N}$ and $b_1, b_2, ... \in \mathbb{N}$ such that $A = \cup_{n\geq 1}\{(a_n, b_n)\}$ (since $\mathbb{N} \times \mathbb{N}$ is countable).

Clearly, $\{(a_n, b_n)\} \in \mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N})$ for all $n \geq 1$. Thus, $A \in \mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N})$ since a σ-algebra is closed under countable unions.

Thus, $\mathcal{P}(\mathbb{N} \times \mathbb{N}) \subseteq \mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N})$.

So $\mathcal{P}(\mathbb{N} \times \mathbb{N}) = \mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N}) = \mathcal{F}_1 \otimes \mathcal{F}_2$ □

(ii) Yes, f is a simple function and the sets associated with each indicator, $\{(n, n) : n \in \mathbb{N}\}$ and $\{(n, n + 1) : n \in \mathbb{N}\}$ are both elements of $\mathcal{F}_1 \otimes \mathcal{F}_2$.

(iii)

$$\begin{aligned}
&\int_{\mathbb{N}\times\mathbb{N}} |f| d\mu_1 \times \mu_2 \\
&= \int_{\mathbb{N}\times\mathbb{N}} \mathbf{I}_{\{(n,n):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) + \mathbf{I}_{\{(n,n+1):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) d\mu_1 \times \mu_2(\omega_1, \omega_2) \\
&= 1 \cdot \mu_1 \times \mu_2(\{(n, n) :\ n \in \mathbb{N}\}) + 1 \cdot \mu_1 \times \mu_2(\{(n, n + 1) :\ n \in \mathbb{N}\}) \\
&= 1 \cdot \infty + 1 \cdot \infty = \infty
\end{aligned}$$

Thus, f is not integrable w.r.t $\mathcal{F}_1 \otimes \mathcal{F}_2$.

(iv)

$$\begin{aligned}
&\int_{\Omega_1} \left[\int_{\Omega_2} f(\omega_1, \omega_2) d\mu_2(\omega_2) \right] d\mu_1(\omega_1) \\
&= \int_{\Omega_1} \left[\int_{\Omega_2} \mathbf{I}_{\{(n,n):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) - \mathbf{I}_{\{(n,n+1):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) d\mu_2(\omega_2) \right] d\mu_1(\omega_1) \\
&= \int_{\Omega_1} \left[\mu_2(\{(\omega_1, n) :\ n = \omega_1\}) - \mu_2(\{(\omega_1, n+1) :\ n = \omega_1 + 1\}) \right] d\mu_1(\omega_1) \\
&= \int_{\Omega_1} 0 d\mu_1(\omega_1) = 0
\end{aligned}$$

$$\begin{aligned}
&\int_{\Omega_2} \left[\int_{\Omega_1} f(\omega_1, \omega_2) d\mu_1(\omega_1) \right] d\mu_2(\omega_2) \\
&= \int_{\Omega_2} \left[\int_{\Omega_1} \mathbf{I}_{\{(n,n):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) - \mathbf{I}_{\{(n,n+1):\, n\in\mathbb{N}\}}(\omega_1, \omega_2) d\mu_1(\omega_1) \right] d\mu_2(\omega_2) \\
&= \int_{\Omega_2} \left[\mu_1(\{(n, \omega_2) :\ n = \omega_2\}) - \mu_1(\{(n, \omega_2) :\ n = \omega_2 - 1\}) \right] d\mu_2(\omega_2) \\
&= \int_{\Omega_1} 0 d\mu_2(\omega_2) = 0
\end{aligned}$$